

APPENDIX 11 – RECURSIVE LOGIC ANALYSIS AND MECHANISM MAPPING

Part 1: Inventory of Recursive Logic Patterns

Chapter 5's Digital Sovereignty section contains D-A-S Assessments across three subsections: six sovereignty dimensions, five claimant types, and six debates. Each assessment identifies how design choices create affordances that structure sovereignty possibilities, and most articulate a recursive pattern describing how these elements cycle through ongoing contestation. Seventeen distinct recursive patterns were identified.

Dimensions

1. Political. Existing state power shapes what technical architectures are mandated or permitted; those architectures determine what actions are possible for users, platforms, and other states; emergent practices and resistances prompt new design interventions and sovereignty assertions. No actor enjoys complete autonomy; the cycle operates through perpetual contestation between territorial jurisdiction and networked architecture.

2. Economic. Corporate ownership structures create extractive affordances, generating resistance; resistance prompts regulatory sovereignty assertions; regulations mandate design changes; design changes redistribute affordances; new affordances enable alternative ownership models; alternatives challenge incumbent sovereignty. No element determines outcomes; all remain contested through ongoing struggle over who controls digital value creation and distribution.

3. Technical. Infrastructure concentration limits alternatives, prompting sovereignty assertions (regulatory mandates or self-hosting adoption); these assertions require technical capacity (design of expertise and systems); new technical capacity enables alternative architectures; alternative architectures redistribute affordances; redistributed affordances enable new sovereignty claims. Yet dependencies persist, ensuring technical sovereignty remains a site of ongoing contestation.

4. Legal. Legal mandates require technical implementation; technical designs structure compliance possibilities; compliance gaps prompt stronger mandates; stronger mandates demand greater technical capacity; capacity requirements favor resourced actors; concentration generates sovereignty concerns; concerns motivate legal intervention. No element independently determines outcomes; all remain contested sites where sovereignty struggles unfold through ongoing negotiation between formal authority and practical capacity.

5. Social. Cultural marginalization shapes whose perspectives inform design; design embeds dominant cultural assumptions; embedded assumptions structure who can participate meaningfully; participation barriers determine whose communities exercise sovereignty; sovereignty inequalities influence future design. Breaking this cycle requires intentional counter-design: Indigenous data governance protocols, community-owned networks, multilingual AI systems, participatory platform development.

6. Environmental. Corporate power shapes infrastructure design; design externalizes environmental costs; communities bearing costs lack decision-making authority; sovereignty deficits enable further extraction; extraction funds political influence maintaining favorable regulations; regulations perpetuate design patterns; patterns structure continued externalization. Twin extractivism operates recursively: data appropriation requires energy-intensive infrastructure; environmental degradation generates data that platforms appropriate; appropriated data trains AI systems demanding more computational resources, accelerating environmental extraction.

Sovereignty Claimants (Patterns 7–11)

7. States. Regulations attempt to constrain corporate power; corporations restructure operations to comply minimally or evade requirements; regulatory inadequacy generates civil society pressure for stronger rules; states strengthen mandates; corporations intensify

lobbying for weaker enforcement; captured mechanisms undermine rules despite formal authority. Community-based claimants enter this cycle as both beneficiaries of regulatory victories and mobilizers of pressure for stronger protections.

8. Corporations. Corporate sovereignty operates recursively through network effects and data accumulation. Each additional user increases platform value, strengthening lock-in affordances that entrench sovereignty. Each transaction generates behavioral data improving effectiveness and competitor advantage. Sovereignty spirals result: market dominance generates data that improves services that attract users that generate more data that strengthens dominance. A secondary cycle: corporate profit funds lobbying weakening regulations; weakened regulations enable continued extraction; extraction generates profits funding further political influence.

9. Academia. Corporate and state actors fund research amplifying perspectives compatible with existing power while marginalizing transformative alternatives; funded research shapes policy debates; resulting regulations perpetuate rather than challenge extraction; perpetuation generates new research questions. These completing cycles either deepen or contest academic capture.

10. Policy Advocacy. High-profile victories attract funding enabling more cases, building expertise improving future strategies; think tanks whose recommendations governments adopt gain credibility attracting consultancy contracts and policymaker audiences; success breeds access breeding further success. Conversely, organizations whose advocacy fails struggle to attract funding, losing capacity and relevance. The ecosystem concentrates power in established organizations, potentially excluding grassroots voices.

11. Community-Based. Design choices encoding participation and collective ownership create affordances for democratic practice, which strengthen sovereignty capacities through exercised agency. Communities governing mesh networks, data commons,

or cooperative platforms develop technical expertise, organizational capacity, and institutional confidence enabling more ambitious design projects. Success attracts participation, generating network effects challenging incumbent extraction models. However, visibility invites co-optation, generating new design challenges: scaling while preserving democratic governance, preventing enclosure through licensing innovations, sustaining volunteer-dependent systems through institutional anchoring. Community sovereignty remains perpetually under construction through continuous design innovation.

Sovereignty Debates (Patterns 12–17)

12. State vs. Corporate Control. Corporate designs afford extraction and lock-in, generating pressure for regulation; state regulations redesign interfaces to afford oversight and competition, triggering corporate resistance and evasion; community alternatives design systems affording democratic governance, requiring supportive legal frameworks states must provide. Each cycle generates new design challenges as actors adapt. Sovereignty remains perpetually contested, negotiated through design choices structuring which actors possess capacity to do what under which constraints.

13. Individual vs. Collective Rights. Individual rights designs afford personal control while undersupplying collective remedies, generating pressure for collective governance; collective designs afford community authority while risking majoritarian control, requiring individual protection mechanisms; Indigenous designs afford relational stewardship while depending on recognition from institutions structured around individual-collective binaries.

14. Localization vs. Free Flow. Surveillance revelations generate pressure for localization; localization fragments services, generating pressure for qualified flow mechanisms; qualified flows prove inadequate when enforcement fails or receiving jurisdictions change practices, generating renewed localization demands. Durable sovereignty

hinges on transfer architectures embedding verifiable safeguards and reciprocal accountability.

15. Technological Determinism vs. Social Construction. Technical elites invoke determinism to restrict governance scope, claiming democratic input threatens system integrity; democratic movements assert constructivism to expand governance scope, claiming technical choices serve values requiring political deliberation. Claims about technical necessity often mask political choices; assertions of determinism themselves constitute political moves restricting sovereignty.

16. Reform vs. Transformation. Reform designs afford oversight but preserve extractive ownership structures, generating pressure for transformation when incremental improvements prove insufficient; transformation designs create affordances but lack scale and resources; reforms then create enabling conditions for transformation; successful transformations justify deeper reforms.

17. Open vs. Closed Systems. Corporate appropriation of open-source creates pressure for protective licenses; restrictive licenses reduce adoption, generating pressure for openness; openness without protection affords extraction; closure without transparency concentrates power. Effective sovereignty emerges from encoding reciprocity in technical and legal architectures, creating affordances for participation while constraining extractive appropriation.

Part 2: Qualitative Mechanism Types from Chapter 4

The qualitative cross-impact matrix characterized each condition-outcome connection through action verbs describing the mechanism at work. Analysis revealed five dominant mechanism types:

Structural mechanisms configure authority and governance architecture.

Governance conditions (GV1–GV6) operate primarily through structural mechanisms,

determining who holds decision-making authority, how authority is distributed, and what governance forms are possible. Primary actions: configure, distribute, structure, govern, establish.

Enabling mechanisms build capacity for action. Technology Practices (TP1–TP6) and Resources (RS1–RS6) conditions operate primarily through enabling mechanisms, providing technical affordances and material inputs that make new forms of action possible. Primary actions: enable, ground, design, build, support, sustain.

Connective mechanisms build relational infrastructure. Partner conditions (PT1–PT5) operate primarily through connective mechanisms, positioning cases across networks, creating relational affordances for cross-scale engagement. Primary actions: connect, position, mediate, bridge, federate.

Protective mechanisms secure gains against appropriation. Regulation conditions (RG1–RG4) and aspects of Governance operate through protective mechanisms, creating safeguards that defend community achievements from enclosure, co-optation, or reversal. Primary actions: protect, secure, defend, preserve, shield.

Legitimizing mechanisms construct external recognition. Purpose (PU1–PU6) and Politics (PO1–PO3) conditions operate through legitimizing mechanisms, grounding transformative efforts in clear values, constructing external recognition, and driving advocacy extending internal achievements into systemic transformation. Primary actions: legitimize, ground, advocate, justify, validate.

These mechanisms vary systematically by outcome group. Democratic Practices outcomes rely primarily on structural and enabling mechanisms. Community Value outcomes rely on protective and enabling mechanisms. Balance outcomes rely on connective and enabling mechanisms. Impact outcomes rely on legitimizing and enabling mechanisms.

Part 3: Convergence Analysis

The comparison reveals systematic convergence between the condition-outcome mechanism types (Chapter 4) and the D-A-S recursive patterns (Chapter 5). The five mechanism types are the micro-level instantiation of the recursive pathways. The table below summarizes the connection between mechanisms, conditions, and recursive logic.

Table 1 Convergence of mechanisms, conditions, and recursive patterns

Mechanism Type	Primary Conditions	Corresponding Recursive Patterns	Convergence Logic
Structural	Governance (GV1–GV6)	Political dimension (#1); States (#7); State vs. Corporate (#12); Reform vs. Transformation (#16)	Governance conditions configure authority distribution. The recursive patterns describe how authority is contested: states mandate designs, corporations resist, communities construct alternatives. Structural mechanisms are the condition-level expression of the political recursive cycle.
Enabling	Technology Practices (TP1–TP6); Resources (RS1–RS6)	Technical dimension (#3); Community-based (#11); Determinism vs. Construction (#15)	Technology and resource conditions create capacity. The recursive patterns describe how technical capacity enables alternative architectures that redistribute affordances. Enabling mechanisms are the condition-level expression of the technical recursive cycle.
Connective	Partners (PT1–PT5)	Individual vs. Collective (#13); Localization vs. Free Flow (#14); Policy Advocacy (#10)	Partner conditions build networks and relationships. The recursive patterns describe how individual and collective governance designs generate ongoing tensions requiring relational navigation. Connective mechanisms are the condition-level expression of the relational recursive cycle.
Protective	Regulation (RG1–RG4); Governance (GV2, GV4)	Legal dimension (#4); Open vs. Closed (#17)	Regulation and ownership conditions secure gains. The recursive patterns describe how legal mandates cycle through implementation, compliance gaps, and strengthened mandates. Protective mechanisms are the condition-level expression of the legal recursive cycle.
Legitimizing	Purpose (PU1–PU6); Politics (PO1–PO3)	Societal/Cultural dimension (#5); Environmental	Purpose and politics conditions construct recognition and drive advocacy. The recursive patterns describe how cultural marginalization

		dimension (#6); Academia (#9)	cycles through design exclusion. Legitimizing mechanisms are the condition-level expression of the epistemic and cultural recursive cycles.
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The convergence is not coincidental. The D-A-S recursive patterns describe, at the level of sovereignty dimensions and political economy, the same dynamics that the qualitative mechanism descriptors capture at the level of individual condition-outcome relationships. When Chapter 4 identifies that Governance conditions "structure" Democratic Practices outcomes through authority configuration, it describes the same dynamic that Chapter 5's Political D-A-S Assessment describes as sovereignty assertions mandating design changes that restructure authority distribution. The condition-outcome mechanisms are how the recursive D-A-S dynamics manifest in community-led digital initiatives specifically.

Three specific implications follow.

First, the five mechanism types can be understood as five modalities through which the D-A-S recursive logic operates: structuring authority (through governance design), enabling capacity (through technical design), connecting across scales (through relational design), protecting gains (through legal-institutional design), and legitimizing claims (through epistemic and political design). Each modality corresponds to specific D-A-S patterns.

Second, the recursive character of the D-A-S patterns explains why conditions do not produce outcomes additively. The configurational logic documented in the case equations reflects the recursive interdependence of the D-A-S cycle: governance conditions structure what technology conditions can enable; technology conditions create affordances that partnership conditions navigate; partnership conditions build networks through which political conditions achieve legitimacy; political conditions reshape the governance terrain on

which the cycle operates. Removing any element disrupts the recursive cycle, explaining why specific condition configurations matter more than the sheer number of conditions present.

Third, the convergence provides the theoretical explanation for Bridge C (Mechanistic): D-A-S reveals how conditions relate to outcomes through multiple recursive pathways. Different conditions operate through different D-A-S logics, and the qualitative mechanism descriptors identify which logic operates in which condition-outcome relationship.

Part 4: Implications for the Agent-Based Model

The model's production rules operationalize the micro-level mechanisms. The model's simulation dynamics operationalize the macro-level recursive patterns. The model thus captures both the mechanisms through which conditions produce outcomes within individual cases and the recursive dynamics through which sovereignty is contested across the case ecosystem. Sensitivity analysis can test whether the five mechanism types operate with different weights across organizational types. Counterfactual scenarios can explore what happens when specific recursive cycles are disrupted. Emergent pattern detection can reveal which recursive cycles are most robust and which are most vulnerable to disruption across 43 cases. The charts the model produces can be understood as formalized descriptions of how specific recursive D-A-S cycles operate across community-led digital initiatives.